AIMS African Institute for Mathematical Sciences RWANDA

Lecture 2: Probability II

Introduction to Probability and Statistics

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Probability on finite sample space

- Our goal is now to measure the chance for an event to occur.
- To proceed we construct a function P that assigns a unique number to each event.
- A natural way:

$$P(A) = \frac{\#A}{\#\Omega}.$$

The function P is such that $0 \leqslant P(A) \leqslant 1$, $P(\Omega) = 1$ and $P(\emptyset) = 0$.

n.b. here Ω is assumed to be finite and countable.

Example

- Random experiment: *Rolling a fair die.*
- The universe for this experiment is:

$$\Omega = \left\{ \text{⚀}, \text{⚁}, \text{⚂}, \text{⚃}, \text{⚄}, \text{⚅} \right\}.$$

  - ➥ $\#\Omega = 6.$
- $A = \{$ *The result is odd* $\} = \left\{ \text{⚀}, \text{⚂}, \text{⚄} \right\}.$
  - ➥ $\#A = 3.$
- Probability for the result to be odd:

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  ➥ $\#A = 3.$
- Probability for the result to be odd:

$$P(A) = \frac{\#A}{\#\Omega} = \frac{3}{6} = \frac{1}{2}.$$

Probability on finite sample spaces

Exercise – A die is rolled twice. Display the sample space for this experiment by listing its elements in full detail. On a diagram, circle the following events:

A: faces are equal.

B: sum of faces is $\leqslant 10$.

C: faces sum to 7.

Compute $P(A)$, $P(B)$ and $P(C)$.

σ-field

Probability will be defined on a collection of events that should be a σ-field. Recall:

Definition

*A collection $\mathscr{F}$ of subsets of Ω is called a σ-**field** if it satisfies the following conditions:*

- $\emptyset \in \mathscr{F}$ *or* $\Omega \in \mathscr{F}$,
- *if* $A \in \mathscr{F}$ *then* $\overline{A} \in \mathscr{F}$.
- *if* $A_1, A_2, \ldots \in \mathscr{F}$ *then* $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$,

Probability measure

Definition

A probability measure P on $(\Omega, \mathscr{F})$ is a function $P : \mathscr{F} \to [0, 1]$ satisfying

1. $P(A) \geqslant 0$, *for all* $A \in \mathscr{F}$.
2. $P(\Omega) = 1$.
3. *If* $A_1, A_2, \ldots$ *is a collection of disjoint members of* $\mathscr{F}$ *(which means that* $A_i \cap A_j = \emptyset$ *for all pairs satisfying* $i \neq j$*), then*

$$P\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} P(A_i).$$

The triple $(\Omega, \mathscr{F}, P)$, is called a ***probability space****.*

The axiom 3 is called the **countable additivity**.

Examples

Example 1:

A coin is tossed once.

We take $\Omega = \{H, T\}$ and $\mathscr{F} = \Big\{\emptyset,\ \{H\},\ \{T\},\ \Omega\Big\}$.

A possible probability measure P is given by:

$$P(\emptyset) = 0\ ,\quad P(H) = p\ ,\quad P(T) = 1 - p\ ,\quad P(\Omega) = 1,$$

where $p \in [0, 1]$.

Examples

Example 2:

Let consider $\Omega = [0, 1]$

$\mathscr{F} = \mathscr{B}\big([0, 1]\big)$, the Borel σ-algebra on $[0, 1]$.

➥ this is the smallest σ-algebra containing all open intervals $(a, b) \subseteq [0, 1]$.

Let A be an interval from $\mathscr{B}\big([0, 1]\big)$

$$P(A) = \text{Upper bound of } A - \text{Lower bound of } A = \text{length of } A.$$

➥ Lebesgue measure.

e.g. $P\big((a, b]\big) = b - a$, $\forall x \in [0, 1], P(\{x\}) = 0$.

Properties

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- For every A, $P(\overline{A}) = 1 - P(A)$.
- Let A and B be two events such that $A \subseteq B$.
  Then $P(A) \leqslant P(B)$.
- For every A, $P(A) \leqslant 1$.
- **Finite Additivity.**
  Let $A_1, \ldots, A_n$ be disjoint events.
  Then $P(A_1 \cup \ldots \cup A_n) = P(A_1) + \cdots + P(A_n)$.

Properties

- **Boole's Inequality.** For any sequence $A_1, A_2, A_3 \dots$

$$P\Big(\bigcup_{i=1}^{\infty} A_i\Big) \leqslant \sum_{i=1}^{\infty} P(A_i).$$

Proof:

Properties

- **Boole's Inequality.** For any sequence $A_1, A_2, A_3 \ldots$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leqslant \sum_{i=1}^{\infty} P(A_i).$$

Proof: Let $B_1 = A_1$ and $B_n = \overline{A}_1 \cap \cdots \cap \overline{A}_{n-1} \cap A_n$.
We have $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} B_n = \emptyset$.
Then, from the definition,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = P\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} P(B_n).$$

But $B_n \subseteq A_n$ therefore $P(B_n) \leqslant P(A_n)$ and the result follows.

Properties

Lemma

Let A and B be two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

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Proof

$A \cup B = (A \cap \overline{B}) \cup (A \cap B) \cup (\overline{A} \cap B)$. (disjoint events)

$P(A \cup B) = P(A \cap \overline{B}) + P(A \cap B) + P(\overline{A} \cap B)$. (additivity)

But $A = (A \cap \overline{B}) \cup (A \cap B)$ and $B = (B \cap \overline{A}) \cup (B \cap A)$.

Therefore $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

and $P(B \cap \overline{A}) = P(B) - P(A \cap B)$.

Then

$$P(A \cup B) = P(A) - \cancel{P(A \cap B)} + \cancel{P(A \cap B)} + P(B) - P(A \cap B).$$

Properties

Exercise

Let A, B, C be 3 events.

Consider $E_1 = A \cap \overline{B} \cap \overline{C}$ and $E_2 = A \cap (B \cup C)$.

We have seen that E_1 and E_2 are disjoint and that $E_1 \cup E_2 = A$.

Suppose that $P(A) = 0.6$, $P(B) = 0.4$, $P(C) = 0.3$, $P(A \cap B) = 0.2$, $P(B \cap C) = 0.1$, $P(A \cap C) = 0.1$ and $P(A \cap B \cap C) = 0.05$.

Calculate $P(E_1)$ and $P(E_2)$

Properties

Solution

- $P(E_2) = P\big(A \cap (B \cup C)\big) = P\big((A \cap B) \cup (A \cap C)\big)$.
  From the previous lemma, we have:

$$\begin{aligned} P\big((A \cap B) \cup (A \cap C)\big) &= P\big(A \cap B\big) + P\big(A \cap C\big) - P(A \cap B \cap C) \\ &= 0.2 + 0.1 - 0.05 = \boxed{0.25} \end{aligned}$$

- We know that $E_1 \cup E_2 = A$ and $E_1 \cap E_2 = \emptyset$ then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) \iff P(A) = P(E_1) + P(E_2)$$

  And, $P(E_1) = P(A) - P(E_2) = 0.6 - 0.25 = \boxed{0.35}$

Properties

Poincaré's theorem:

Let $A_1, \ldots, A_n$ be events. Then

$$\begin{aligned} P(A_1 \cup A_2 \cup \ldots \cup A_n) &= \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) \\ &+ \sum_{i<j<k} P(A_i \cup A_j \cap A_k) - \cdots \\ &+ (-1)^{n+1} P(A_1 \cap \ldots \cap A_n). \end{aligned}$$

Conditional Probability

Consider two events A and B. Suppose that the event B occurs.

What can we say about the chance of occurence for event A?

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Definition

If $P(B) > 0$ the conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- **Remark:** $P(A \cap B) = P(A \mid B)P(B)$.

Conditional Probability

Lemma

For any events A and B such that $0 < P(B) < 1$,

$$P(A) = P(A \mid B)P(B) + P(A \mid \overline{B})P(\overline{B}).$$

Proof

Conditional Probability

Lemma

For any events A and B such that $0 < P(B) < 1$,

$$P(A) = P(A \mid B)P(B) + P(A \mid \overline{B})P(\overline{B}).$$

Proof

$A = (A \cap B) \cup (A \cap \overline{B})$. $A \cap B$ and $A \cap \overline{B}$ are disjoint.

Then $P(A) = P(A \cap B) + (A \cap \overline{B})$.

But $P(A \cap B) = P(A \mid B)P(B)$ and $P(A \cap \overline{B}) = P(A \mid \overline{B})P(\overline{B})$.

Therefore

$$P(A) = P(A \mid B)P(B) + P(A \mid \overline{B})P(\overline{B}).$$

Conditional Probability

Generalization:

The Law of Total Probability

Let $B_1, \ldots, B_n$ be a partition of Ω. Then, for any event A,

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i).$$

Bayes' Theorem

Let $B_1, \ldots, B_n$ be a partition of Ω such that $P(B_i) > 0$ for each i. If $P(A) > 0$ then, for each $i = 1, \ldots, n$,

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}.$$

Conditional Probability

Exercise

Larry divides his email into three categories A_1= *"spam"*, A_2= *"low priority"* and A_3= *"high priority"*.

From previous experience, he finds that $P(A_1) = 0.7$, $P(A_2) = 0.2$ and $P(A_3) = 0.1$.

Let B be the event that the email contains the word *"free"*.

From previous experience, Larry knows that $P(B \mid A_1) = 0.9$, $P(B \mid A_2) = 0.01$ and $P(B \mid A_3) = 0.01$.

Larry receives an email with the word *"free"*.

What is the probability that it is *"spam"* ?

Conditional Probability

Solution

By definition: $P(A_1 \mid B) = \dfrac{P(A_1 \cap B)}{P(B)}$.

$P(A_1 \cap B) = P(B \mid A_1)P(A_1) = 0.9 \times 0.7 = 0.63.$

By the Law of Total Probability:

$$\begin{aligned} P(B) &= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + P(B \mid A_3)P(A_3) \\ &= 0.9 \times 0.7 + 0.01 \times 0.2 + 0.01 \times 0.1 = 0.633 \end{aligned}$$

- Conclusion:

$$P(A_1 \mid B) = \frac{0.63}{0.633} = 0.9952$$

Independence

The conditional probability caracterizes the fact the occurence of an event B modifies the chance of occurence of A. When it is not the case that is to say the probability remains unchanged $P(A \mid B) = P(A)$, we said that A and B are independent.

Definition

Two events are said to be independent if

$$P(A \cap B) = P(A)P(B).$$

More generally, a family $\{A_i : i \in I\}$ *is called independent if*

$$P\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} P(A_i)$$

for all finite subsets J *of* I.

Independence

Exercise

Toss a fair coin n times. Let A be the event *"At least one head occurs"*.
Let T_j be the event that tails occurs on the j^{th} toss.
Use T_j, $j = 1, \ldots, n$ and their properties to compute the probability of A?

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Solution:
$P(A) = 1 - P(\overline{A}) = 1 - P(\textit{no heads at all}) = 1 - P(\textit{all tails})$.
The event $\{\textit{all tails}\}$ *is* $T_1 \cap T_2 \cap \ldots \cap T_n$ *that is tail occurs on the first toss and on the second toss and so on until the* n^{th} *toss.*
The tosses are independent thus: $P(T_1 \cap T_2 \cap \ldots \cap T_n) = \prod_{i=1}^{n} P(T_j)$.
$T_j = \Big\{H \textit{ or } T, (j-1) \textit{ times}, \boldsymbol{T}, H \textit{ or } T, (n-j) \textit{ times}\Big\}$.
$\# T_j = 2^{j-1} \times 2^{n-j} = 2^{n-1}$ *and* $\#\Omega = 2^n$, *then* $P(T_j) = 1/2$ *and:*

$$P(A) = 1 - \frac{1}{2^n}.$$

Independence

Exercise

Two people are taken turns trying to sink a basketball into a net. Person 1 succeeds with probability $1/3$ while person 2 succeeds with probability $1/4$.
What is the probability that person 1 succeeds before person 2?

Hint: Consider the possible sequences of successive games with F for the event "the player fails" and S for "the player succeeds".

Independence

Solution

Suppose that Person 1 starts the game before Person 2.
Denote E the event of interest.

Person	1	2	1	2	1	2	...	1	2	1	...	*Probability*
First game	S											$1/3$
Second game	F	F	S									$2/3 \times 3/4 \times 1/3$
Third game	F	F	F	F	S							$2/3 \times 3/4 \times 2/3 \times 3/4 \times 1/3$
							$\vdots$					$\vdots$
k^{th} game	F	F	F	F	F		...	F	F	S		$(2/3 \times 3/4)^{k-1} \times 1/3$
$\vdots$												$\vdots$

$$P(E) = \sum_{k=1}^{+\infty} (2/3 \times 3/4)^{k-1} \times 1/3 = \sum_{k=1}^{+\infty} (1/2)^{k-1} \times 1/3 = \frac{1}{3}\left(\frac{1}{1-1/2}\right) = 2/3.$$